

An Exact Fold Condition for Mass-Balanced Models of Protein Quality Control

Kiran Boggavarapu^{1*}

^{1*}Department of Chemistry and Physics, McNeese State University,
Lake Charles, LA, 70609, USA, ORCID 0000-0003-0751-6459.

Corresponding author(s). E-mail(s): kiran@mcneese.edu;

Abstract

A cell clearing misfolded protein with finite pools of chaperones and proteases tolerates damage influx only up to a threshold, beyond which no low-burden steady state exists. It is normally located by continuation, one parameter set at a time. We show it is algebraic. For any clearance network in which the damage influx is state-independent and mass balance accounts for all outflow, the Jacobian determinant factors as $\det J = -(\nabla \mathbf{R} \times \nabla \mathbf{G})$, where $\mathbf{R}$ is total removal flux and $\mathbf{G}$ the aggregate nullcline field. A fold is therefore exactly a constrained critical point of removal on the aggregate nullcline, with critical influx $j_{crit} = \mathbf{R}(\mathbf{u}^*, \mathbf{a}^*)$; the converse holds under three genericity conditions and a fourth that follows from the first. The identity generalises to arbitrary state dimension and survives growth dilution and self-damaging clearance machinery. At the fold the machinery delivers 8.3% of the flux its pools could support, so the capacity ceiling overestimates the tolerable influx about twelvefold under the stipulated catalytic closure and 2.8-fold under the standard bound-complex form. Two results bound it: the fold equations return a candidate set with more than one element in 9% of a kinetic ensemble, and in 3.9% a Hopf bifurcation precedes the fold, which in 47 of those 108 opens a band of influx carrying sustained oscillation. For aggregation asymmetry in dividing *Escherichia coli*, the framework yields a falsifiable requirement on the old-pole aggregate load, indexed by the share in the visible polar focus.

Keywords: proteostasis, saddle-node bifurcation, conserved resource, clearance network, growth dilution, Hopf bifurcation

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